

Royal Canadian Air Cadets
110 Black Hawk Squadron
Proficiency Level 2 Aviation Exam
March 25, 2024
Duration: 60 minutes

Rank: _____

Surname: _____

Given Name: _____

CIN: _____

Exam Information & Instructions: **Please read carefully**

This test is based on multiple-choice questions. There are **x** questions. Ensure you have all **x** pages attached. All questions carry equal weight. On the front of the Exam Sheet, ensure that you write your rank, last name, first name and cadet information number. On the Exam Sheet, you can use either pencil or pen to circle the most correct answer.

Ensure you **DO NOT** have any electronic devices (phone, smartwatch, headphones) on your person and are silenced and in your bag. Any offense will lead to a deduction of marks.

Only items allowed on your desk are your pen/pencil, eraser, transparent water bottle or pencil case. Everything else (opaque containers, wedge/beret) are to be put in your bags.

Only circle **ONE** answer for each question, no marks will be given for questions with multiple answers selected. Cleanly erase any answer you wish to change to ensure there is only **ONE** clear answer. Circle in each case the best answer out of the alternatives given.

You should attempt all questions.

Good Luck!

PO 231 – EXPLAIN PRINCIPLES OF FLIGHT

231.01 - Identify the 4 forces that act upon an aircraft

- Aircraft gains energy gained as altitude increases
- Glider experiences drag when descending
- Glider converts potential energy into thrust by acting upon passing air
- Glider's wings convert energy from descent to lift
- Equilibrium (thrust=drag, lift=weight)

231.02 - Describe the production of lift by an aircraft wing

- Air's fluidity
- Newton's 3 laws of motion
- Bernoulli's Principle
- Angle of Attack

231.03 - Describe the Types of Drag that Act Upon an Aircraft

- How drag can be useful
- Parasite drag (form drag, skin friction)
- Induced drag (wing tip vortices)

231.04 - Describe the Axial Movements of an Aircraft

- 3 axes (longitudinal, lateral, vertical)
- Centre of gravity
- 3 axial movements (roll, pitch, yaw)

231.05 - Describe Aircraft Control Surfaces

- Empennage (horizontal and vertical stabilizer, rudder, elevator)
- Wing control surfaces (ailerons, wing flaps)
- Trim tabs

PO 232 – IDENTIFY CHARACTERISTICS OF PISTON-POWERED AIRCRAFT

232.01 - Identify Types of Aircraft Engines

- Propulsion engine types (rockets, piston-power, gas turbine engines)
- ~~Gas turbine variations (turbojet, turboprop, turboshaft, turbofan)~~
- Internal combustion
- Oxidation for piston power engines
- ~~Engine applications in various aircrafts~~

232.02 - ID the Components of Piston- Powered Internal Combustion Engines

- Major components (cylinder, piston, connecting rod, crankshaft, camshaft, distributor, carburetor, oil sump)
- Rotary and radial engines
- Aircraft engine arrangements (horizontally opposed, in line)

232.03 - Explain the Cycles of a Four-Stroke Piston- Powered Engine

- Cycle (intake, compression, stroke, exhaust)
- Valves and camshafts
- Electric spark timing

232.04 - Recognize the Functions of Oil in a Four-Stroke Piston- Powered Engine

- Oil functions (lubricate, seal, cool, flush)

~~PO 240 – PARTICIPATE IN AEROSPACE ACTIVITIES~~

~~240.01 – Explore Current Advancements in Aerospace Technology~~

~~240.03 – Identify parts of a rocket~~

~~PO 260 – PARTICIPATE IN AERODROME OPERATIONS ACTIVITIES~~

260.01 - Explain Aspects of Air Traffic Control

- ATC's role
- RADAR
- NORDO (no light signals)

260.02 - Identify Aspects of Basic Aerodrome Operations

- Aerodrome operations (air traffic control, ground control, airport Maintenance)
- Ground Controllers
- Ground Facilities/Services (runway maintenance, runway lighting, baggage handling, fuel storage, de-icing, anti-icing)
- CATSA

PRACTICE EXAM QUESTIONS

1. A glider is attempting to maximize its glide ratio while descending. Atmospheric conditions are stable, and no external lift is available. Which of the following actions will most effectively optimize the glide ratio?
 - A. Increasing its descent angle to gain more forward thrust.
 - B. Adjusting its angle of attack to balance lift and drag.
 - C. Increasing its weight to minimize gravitational pull.
 - D. Increasing drag to maintain a slower descent.
2. A team of engineers is designing a new glider to achieve the highest possible glide ratio. They must prioritize trade-offs between wing size, weight, and drag. Which of the following design choices is most likely to maximize glide ratio?
 - A. Small, lightweight wings with minimal surface area to reduce drag.
 - B. Large, wide wings optimized for maximum lift and minimum drag.
 - C. Lightweight fuselage with moderate wings to balance lift and drag.
 - D. Large, heavy wings to counteract gravitational pull and improve lift.
3. A pilot flying a glider detects an area of atmospheric lift and maneuvers to enter it. After spending several minutes in the lift, the glider exits with a higher altitude than it started. What physical principles best explain this phenomenon?
 - A. Conservation of energy and Bernoulli's principle.
 - B. Newton's third law of motion and atmospheric pressure differences.
 - C. Conservation of energy and the upward force from rising air.
 - D. Bernoulli's principle and drag reduction from streamlined design.
4. An aircraft wing generates lift primarily because:
 - A. The top surface of the wing decreases airspeed, causing higher pressure above the wing.
 - B. The bottom surface of the wing increases airspeed, causing lower pressure below the wing.
 - C. The top surface of the wing increases airspeed, lowering pressure above the wing, while the bottom increases pressure.
 - D. The entire wing reduces drag to allow for smooth forward motion.
5. During pre-flight checks, a mechanic identifies ice buildup on the wings. How does this condition primarily affect the aircraft's drag profile?
 - A. Increases induced drag by altering the wing's angle of attack
 - B. Decreases both form and interference drag through surface pressure
 - C. Decreases form drag by adding weight and destabilizing the aircraft
 - D. Increases parasite drag through increased skin friction and disrupted airflow

6. An engineer proposes a new wing design with a high aspect ratio. How would this modification most directly impact the aircraft's performance?
- A. Increase parasite drag by enlarging the wing surface area.
 - B. Reduce induced drag by distributing lift more efficiently.
 - C. Deploy boundary layer control systems to minimize turbulence.
 - D. Prioritize form drag reduction through aerodynamic shaping.
7. During a coordinated turn, a pilot simultaneously applies aileron and rudder inputs. Which axes and associated movements are primarily involved in this maneuver?
- A. Longitudinal axis (roll) and vertical axis (yaw).
 - B. Lateral axis (pitch) and vertical axis (yaw).
 - C. Longitudinal axis (roll) and lateral axis (pitch).
8. While cruising at high altitude, a pilot notices the aircraft tends to pitch upward uncontrollably. Which control surface adjustment would best mitigate this issue without constant manual input?
- A. Extending the flaps to redistribute lift across the wings.
 - B. Adjusting the elevator trim tab to stabilize the horizontal stabilizer.
 - C. Applying full rudder to counteract the pitch moment.
 - D. Retracting the ailerons to minimize lift.
9. A mechanic troubleshooting a broken engine discovers the distributor malfunctioned. Which component's failure most directly explains the broken engine across multiple cylinders?
- A. Spark Plugs – Inconsistent ignition timing.
 - B. Camshaft – Disrupted valve operation.
 - C. Carburetor – Improper fuel-air mixing.
 - D. Connecting Rod – Uneven piston motion.
10. During a routine inspection, an engineer notices metallic particles in the oil filter of a piston-powered engine. Which function of oil is most directly responsible for preventing engine damage from this debris?
- A. Lubricating
 - B. Cooling
 - C. Flushing
 - D. Sealing

1. B
2. B
3. C
4. C
5. D
6. B
7. A
8. B
9. A
10. C